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GEOAI FUSION 2026

Module 4 Session 2: Telecommunication Network Coverage Analysis

AGAIF 2026 Boot Camp 1
Foundation of GeoAI

Thursday 9 July 2026

Delivered by:
Assoc. Prof. Ts. Dr. Wahidah Hashim
Universiti Tenaga Nasional (UNITEN)

Learning Outcomes

By the end of this session, participants should be able to:

1. Explain the concept of telecom network coverage.
2. Visualize network coverage using GIS.
3. Identify areas with poor or no signal.
4. Estimate population coverage using geospatial data.

What is Network Coverage?

Network coverage refers to the geographical area where users can access a communication service with acceptable signal quality.

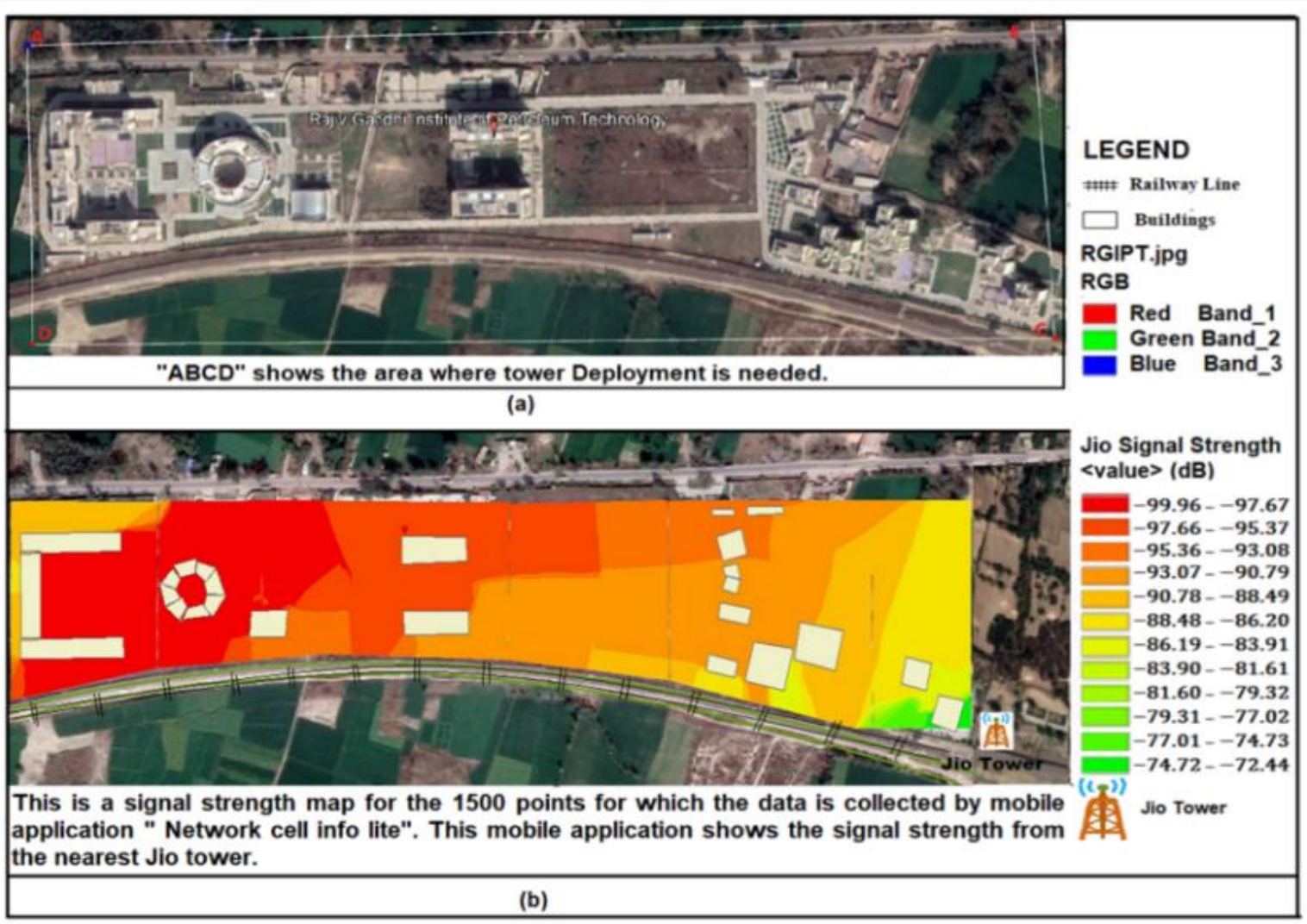
Coverage depends on:

1. Base station location
2. Terrain
3. Buildings
4. Vegetation
5. Distance
6. Frequency

Big
Towers
Bring
Very
Distant
Frequencies



Cellular tower with coloured coverage zones



Open Access Article

A Novel Method to Determine the Optimal Location for a Cellular Tower by Using LiDAR Data

by Shruti Bharadwaj [1,†], Rakesh Dubey [1,†], Md Iltaf Zafar [1,†], Saurabh Kr Tiwary [1], Rashid Aziz Faridi [2] and Susham Biswas [1,*]

[1] Department of Computer Science and Engineering, Rajiv Gandhi Institute of Petroleum Technology, Amethi 229304, India
[2] Department of Geography, Aligarh Muslim University, Aligarh 202001, India
* Author to whom correspondence should be addressed.
† These authors contributed equally to this work.

Why Coverage Analysis Matters

Good network coverage enables:

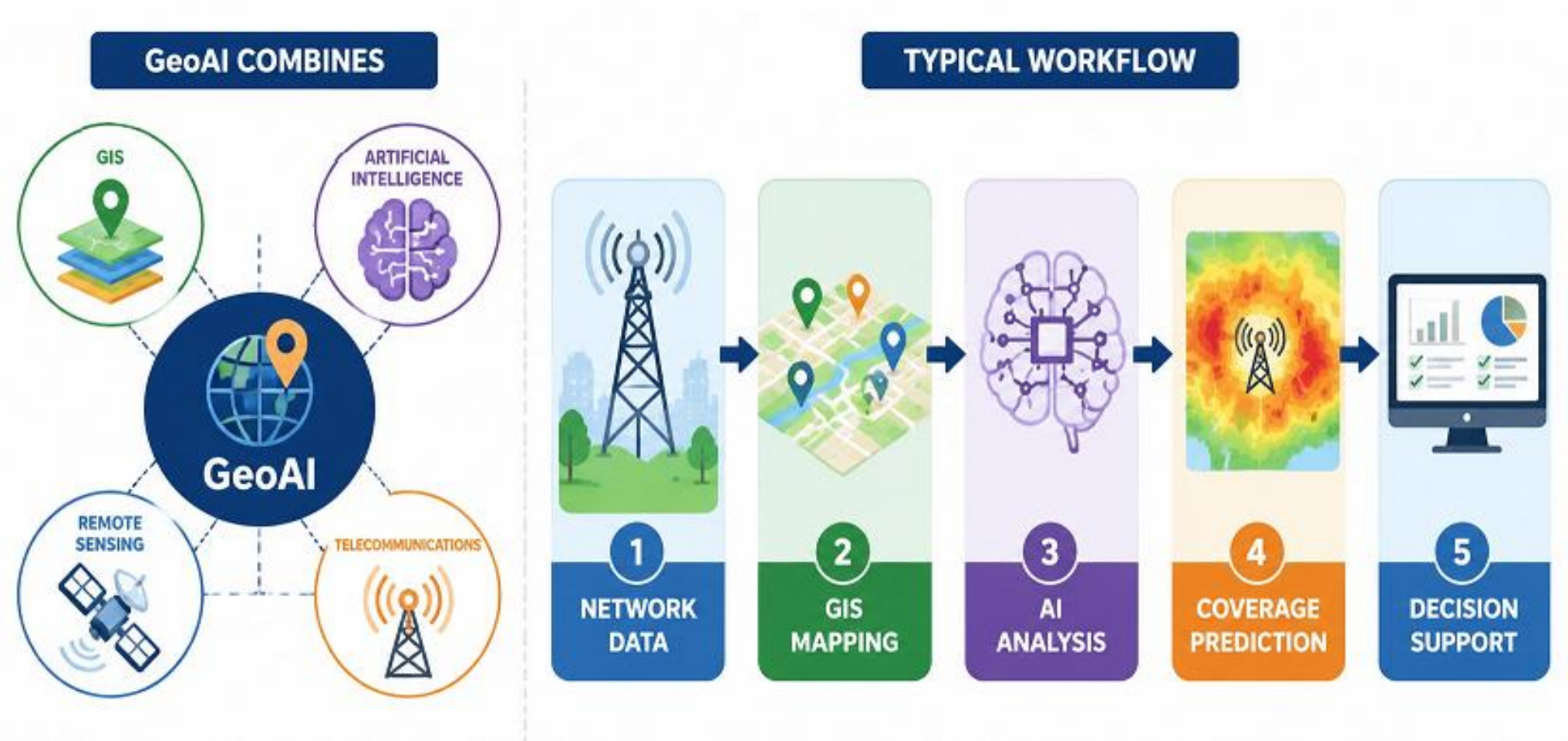
- Reliable communication
- Emergency response
- Smart cities
- IoT applications
- Remote learning
- Digital inclusion

Poor coverage leads to:

- Dropped calls
- Slow Internet
- Digital divide
- Limited access to services



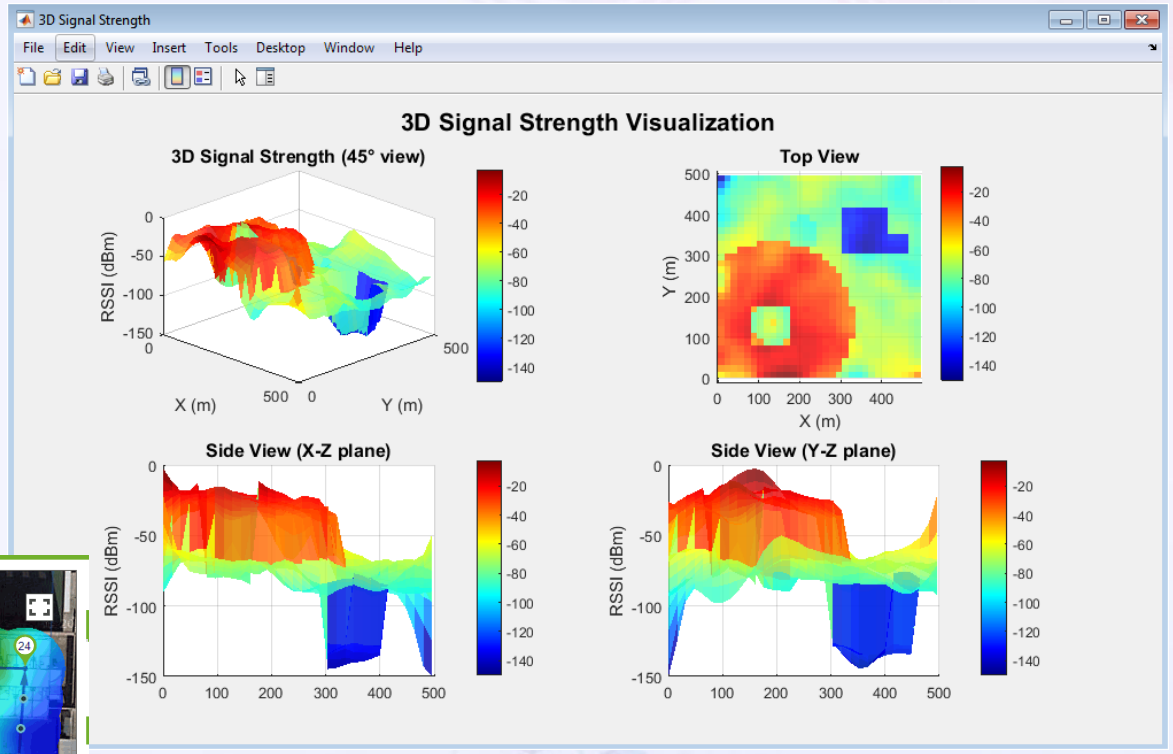
GeoAI for Coverage Mapping

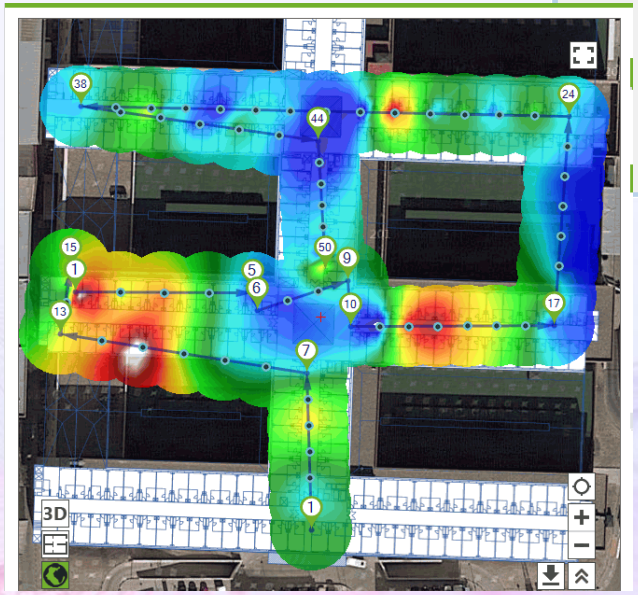


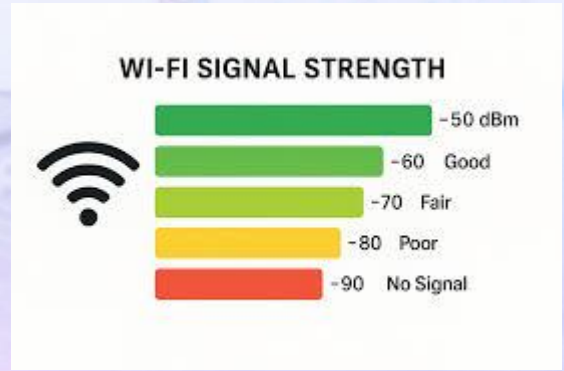
Signal Strength Visualisation

Common visualisation methods:

1. Heat Maps
2. Colour Gradients
3. Contour Maps
4. 3D Surface Maps







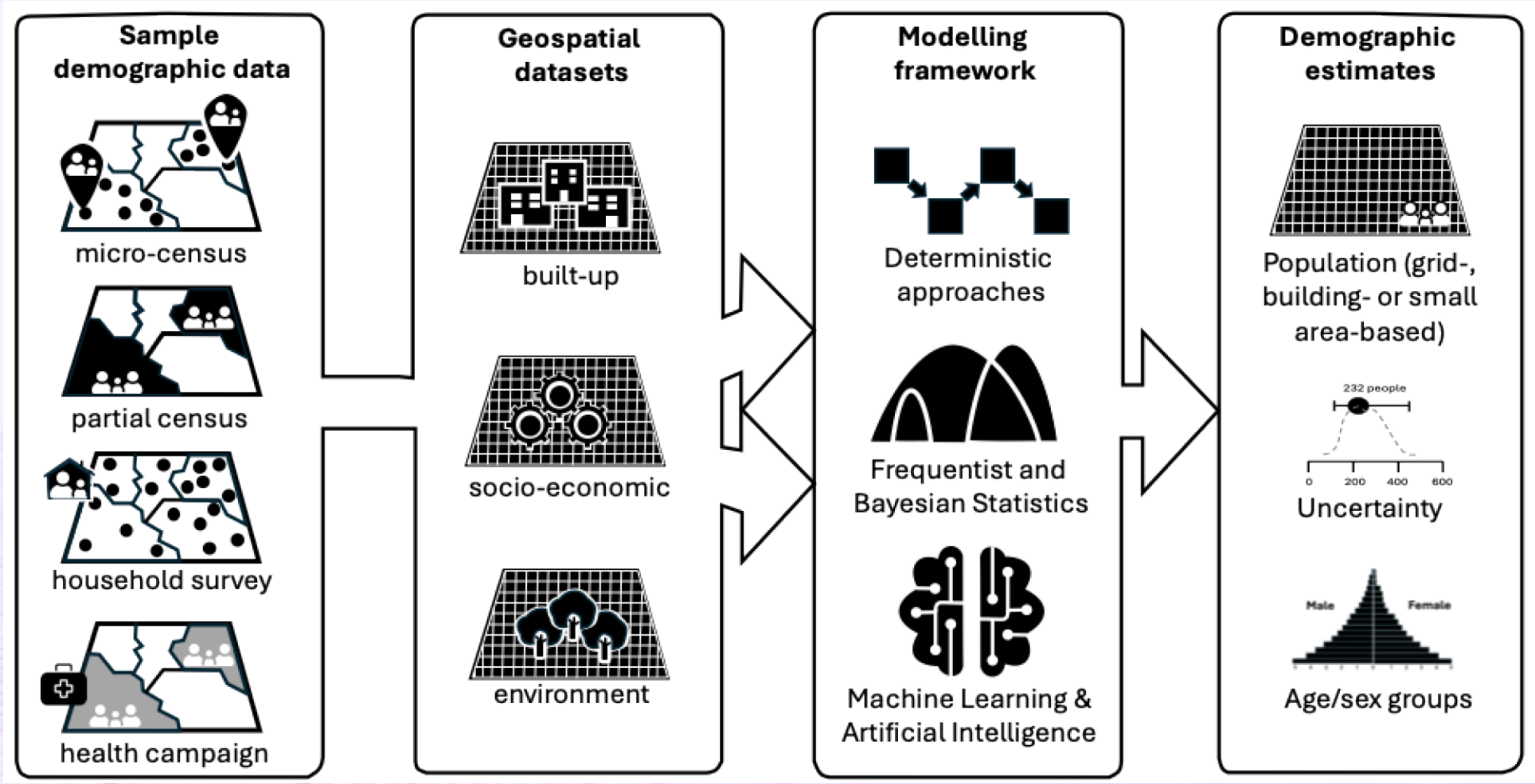
Blind Spot Identification



Population Coverage Estimation

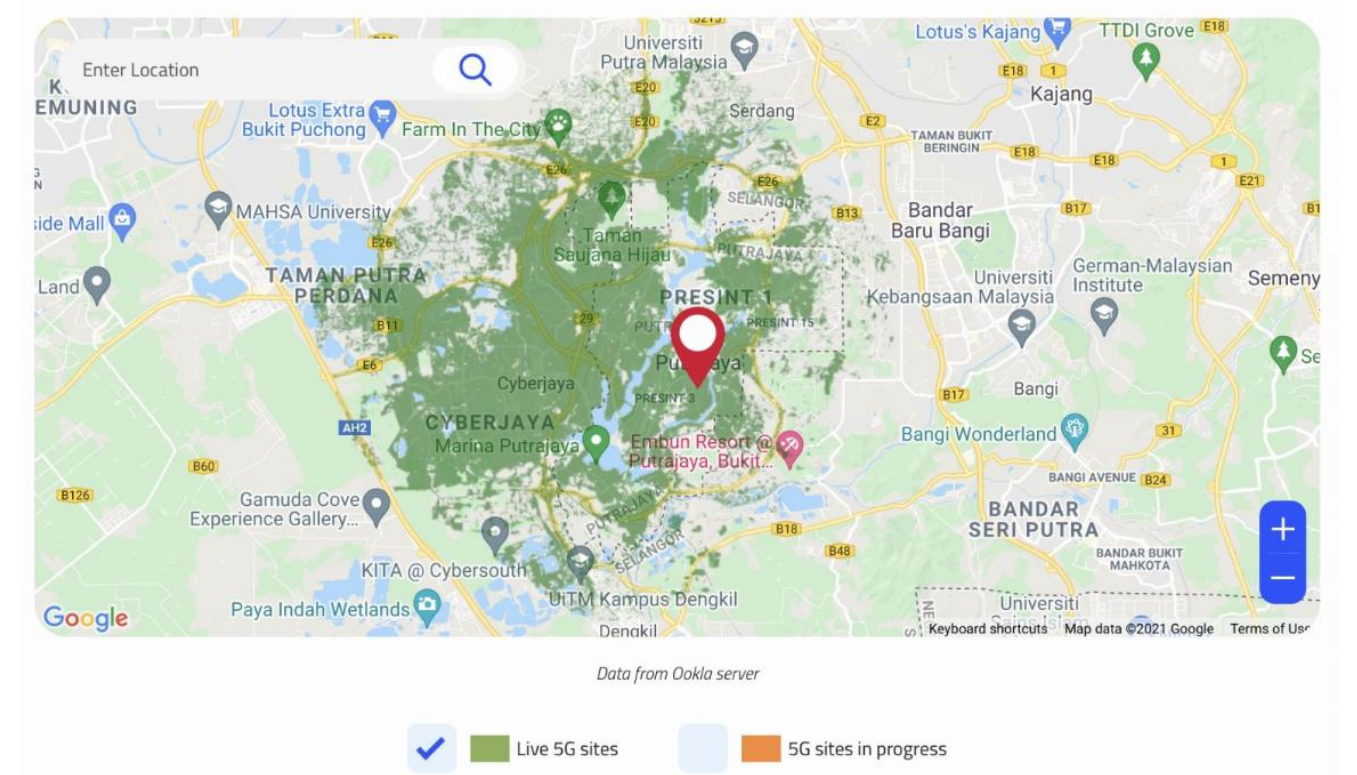
Advances in Small Area Population Estimation in the Absence of National Census Data

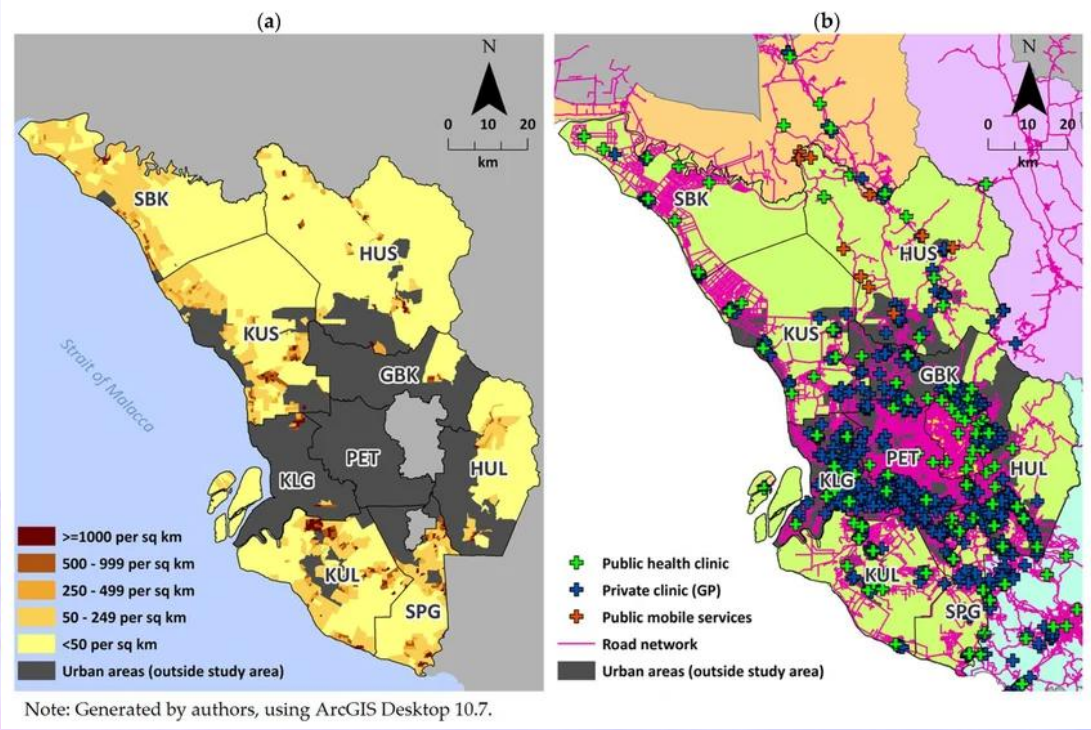
Attila N. Lazar *, Gianluca Boo, Heather R. Chamberlain, Chibuzor



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Population Coverage Estimation





Note: Generated by authors, using ArcGIS Desktop 10.7.

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Population Coverage Estimation Formula

The Core Formula

$$P_{covered} = \sum_{i=1}^{n} (Area_i \times Density_i \times C_i)$$

Where:

- $P_{covered}$: Estimated population with network coverage.
- i: Individual geographic cell, grid, or administrative boundary.
- $Area_i$: The geographical area (in km²) of cell i.
- $Density_i$: The population density of cell i (people per km²).
- C_i: The network coverage factor for cell i. (Assigned as 1 if the signal strength is above your minimum quality threshold, or 0 if it is a "dead zone" or falls below the threshold). AL Neelain University

Core GeoAI Telecom Use Cases

- Infrastructure Optimization: Integrates 3D terrain models, clutter/vegetation data, and building heights to simulate line-of-sight and signal propagation for 4G/5G.
- Micro-Population Modeling: Uses daytime and nighttime population models to capture human mobility, prioritizing bandwidth allocation where people actually congregate.
- Digital Equity Initiatives: Helps policymakers identify localized "digital divides" to efficiently allocate state subsidies for rural and low-density broadband expansion.
- Automated Site Suitability: Uses deep learning and predictive algorithms to evaluate flat terrain, existing power access, and zoning requirements for new tower placements.

ASEAN Case Studies

Examples:

- 📍 Rural broadband expansion
- 📍 Flood emergency communication
- 📍 Smart agriculture connectivity
- 📍 Island connectivity
- 📍 Forest monitoring



Discuss how GeoAI supports network planning in diverse geographical environments.

Key Takeaways

- ✓ Coverage analysis supports better network planning.
- ✓ GIS helps visualise telecom performance.
- ✓ GeoAI identifies blind spots and predicts coverage.
- ✓ Population analysis ensures equitable digital access.

Good network planning begins with good spatial analysis.

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THANK YOU

Bootcamp and Hackathon for ASEAN Geospatial Innovation 2026